

DIGITAL FORENSICS INVESTIGATION REPORT

Case File; Horcrux.ad1

BY

Iwuji Chikaodi Edith

1. CASE OVERVIEW

This report documents the digital forensic investigation conducted on a forensic image file named Horcrux.ad1. The image was acquired using AccessData FTK Imager 4.2.1.4 and contains a custom content capture of selected directories from a Windows machine. The investigation aims to answer 17 specific questions relating to the system configuration, user activity, communications, and file access history of the machine's administrator.

Image File: Horcrux.ad1

Image Format: AD1 (Custom Content Image)

MD5 Checksum: 562ea531434f34bc202b349b3d5f8051

SHA1 Checksum: 6f23af0de52b19be2f12ee148dc2384f535fd781

Acquisition Date: Tuesday, May 11, 2021

Examiner: Minerva

2. TOOLS USED

The following forensic tools were used during the investigation:

Tool	Version	Purpose
FTK Imager	4.7.3.81	Browsing the forensic image file system and exporting registry hives and browser history files
Registry Explorer	v2026.5.0	Reading exported Windows registry hive files (SOFTWARE, SYSTEM, SAM) to extract system information
DB Browser for SQLite	3.13.1	Reading Chrome browser history database to find visited URLs and browser version

SysTools OST Viewer	v5.0	Opening and reading Karen's Outlook email database (.ost file) to view email communications
---------------------	------	---

3. EVIDENCED FILES EXAMINED

The following files were exported from the forensic image and examined:

File	Location in Image	Information Extracted
SOFTWARE (Registry Hive)	Partition 2 > Windows > System32 > config	OS build number
SYSTEM (Registry Hive)	Partition 2 > Windows > System32 > config	Hostname, Timezone, Mounted drive letters
SAM (Registry Hive)	Partition 2 > Windows > System32 > config	User account details, password last changed
History (Chrome SQLite DB)	Partition 2 > Users > Karen > AppData > Local > Google > Chrome > User Data > Default	Browsing history, Skype download URL, Alpaca website
klovespizza@outlook.co m. ost	Partition 2 > Users > Karen > AppData > Local > Microsoft > Outlook	All email communications

4. INVESTIGATION FINDINGS

This section presents each question with the answer found and the exact steps taken to find it.

4.1 Findings from FTK Imager

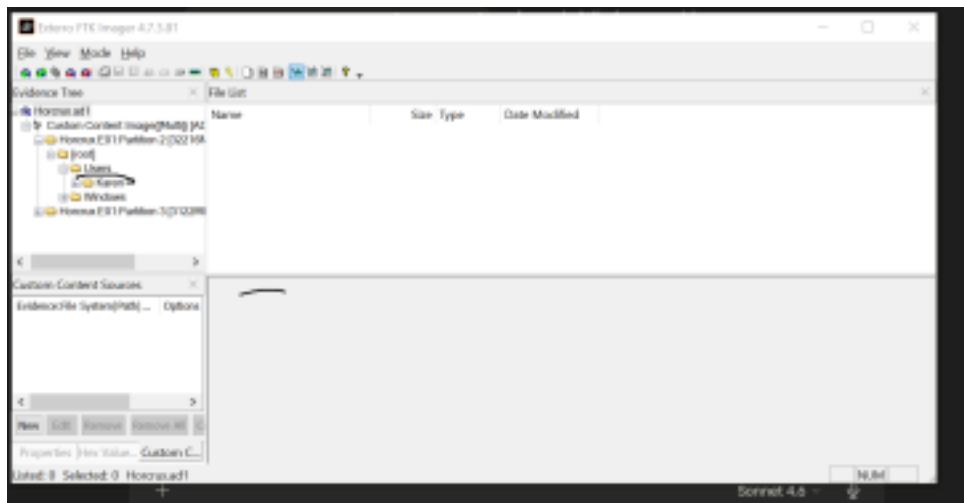
FTK Imager was used to browse the forensic image file system. The image was loaded by clicking File > Add Evidence Item > Image File and selecting Horcrux.ad1.

Q1 - Administrator Username

Question: What is the administrator's username?

Answer: Karen

Method: After loading the image in FTK Imager, the Evidence Tree was expanded by navigating to Partition 2 [NONAME-NTFS] > [root] > Users. The only user folder present was named Karen, confirming she is the administrator of the machine.



Q4 - Messaging Application

Question: A messaging application was used to communicate with a fellow Alpaca enthusiast. What is the name of the software?

Answer: Skype

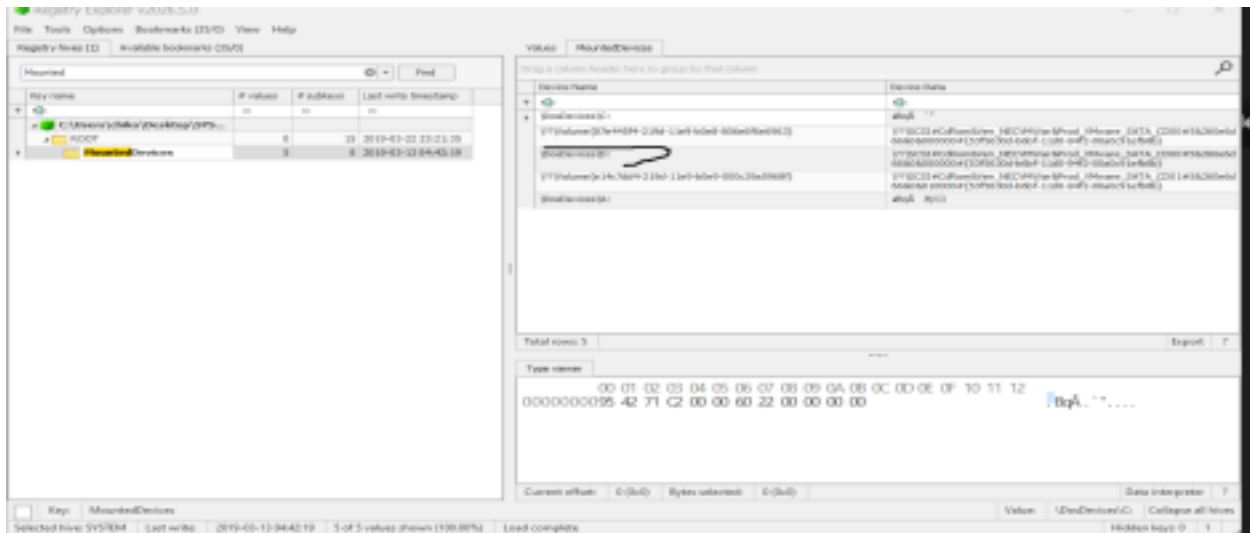
Method: The third partition (Partition 3 - PacaLady) was browsed in FTK Imager. Inside the [root] directory, the file Skype-8.41.0.54.exe was found, which is the Skype installer. This confirms Skype was downloaded and installed on the machine. This finding was further corroborated by the Chrome browser history which showed Karen visited <https://www.skype.com/en/get-skype/> to download Skype.



Q11 - Second Partition Letter

Question: There was a second partition on the drive. What is the letter assigned to it? **Answer:** D

Method: In the SYSTEM registry hive (loaded in Registry Explorer), the MountedDevices key was located by searching for 'MountedDevices'. The right panel displayed all drive letters assigned to partitions. The entry \DosDevices\D: was found mapped to the second partition (PacaLady), confirming the letter D was assigned to it.



4.2 Findings from Registry Explorer

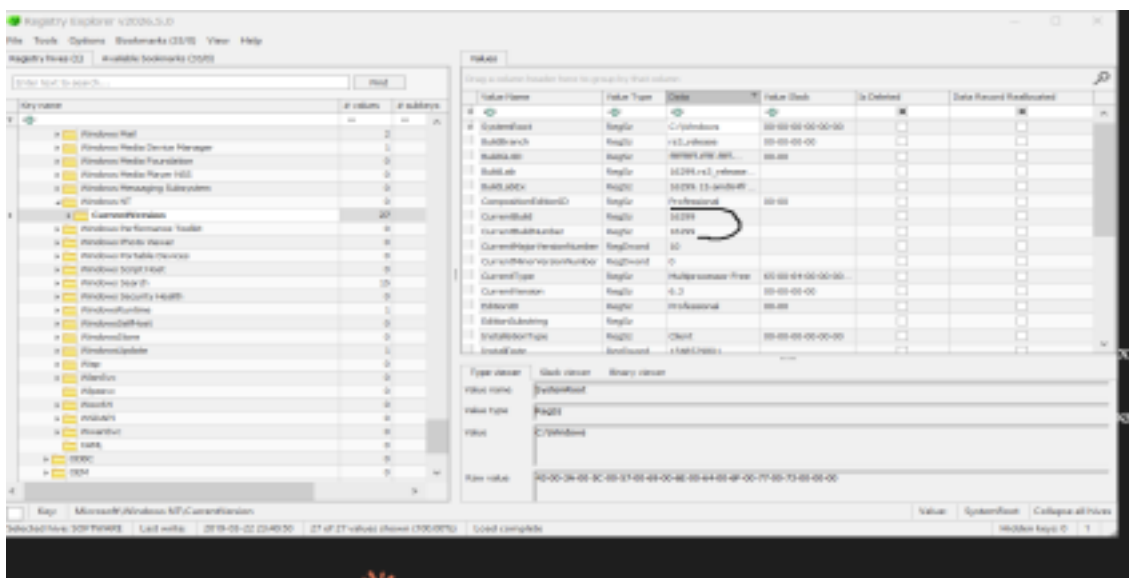
Registry Explorer was used to read the three registry hive files exported from the image. Each hive file was loaded using File > Load Hive.

Q2 - OS Build Number (from SOFTWARE hive)

Question: What is the OS's build number?

Answer: 16299

Method: The SOFTWARE registry hive was loaded. Navigation path: ROOT > Microsoft > Windows NT > CurrentVersion. The value CurrentBuildNumber in the right panel showed 16299. Additional context: BuildBranch showed rs3_release and EditionID showed Professional, indicating this is Windows 10 Pro.

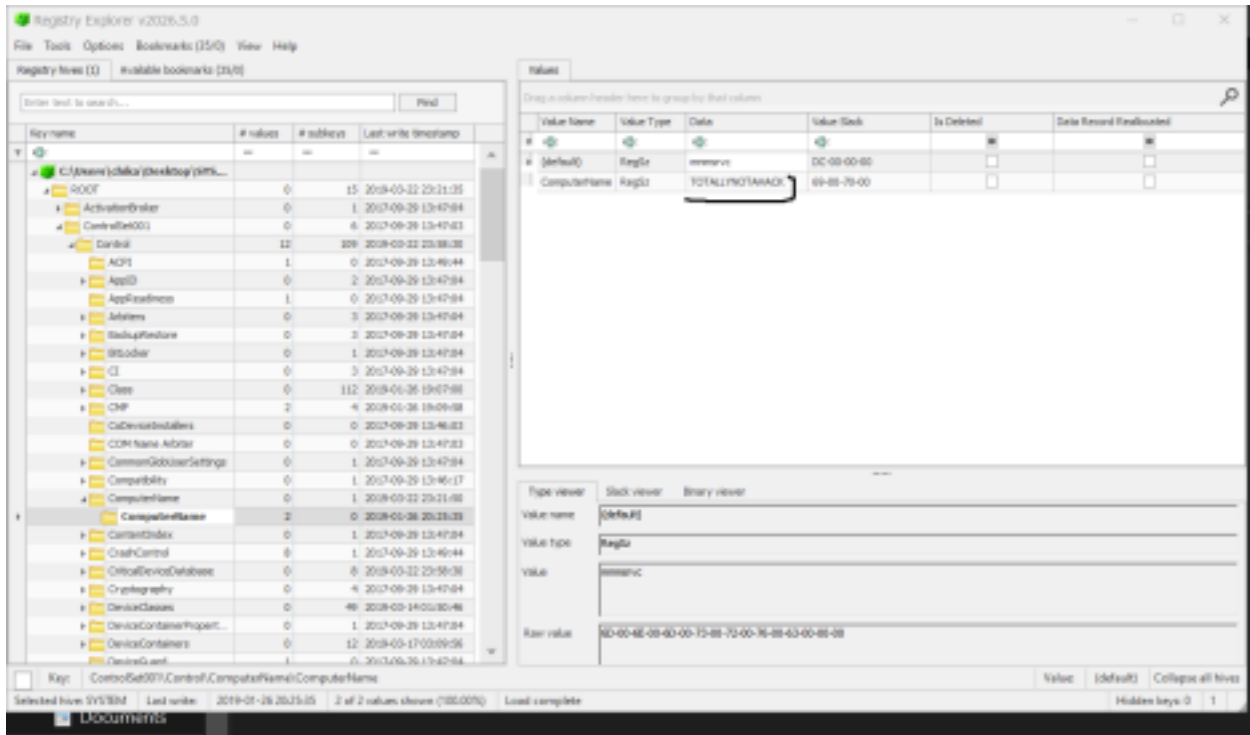


Q3 - Hostname (from SYSTEM hive)

Question: What is the hostname of the computer?

Answer: TOTALLYNOTAHACK

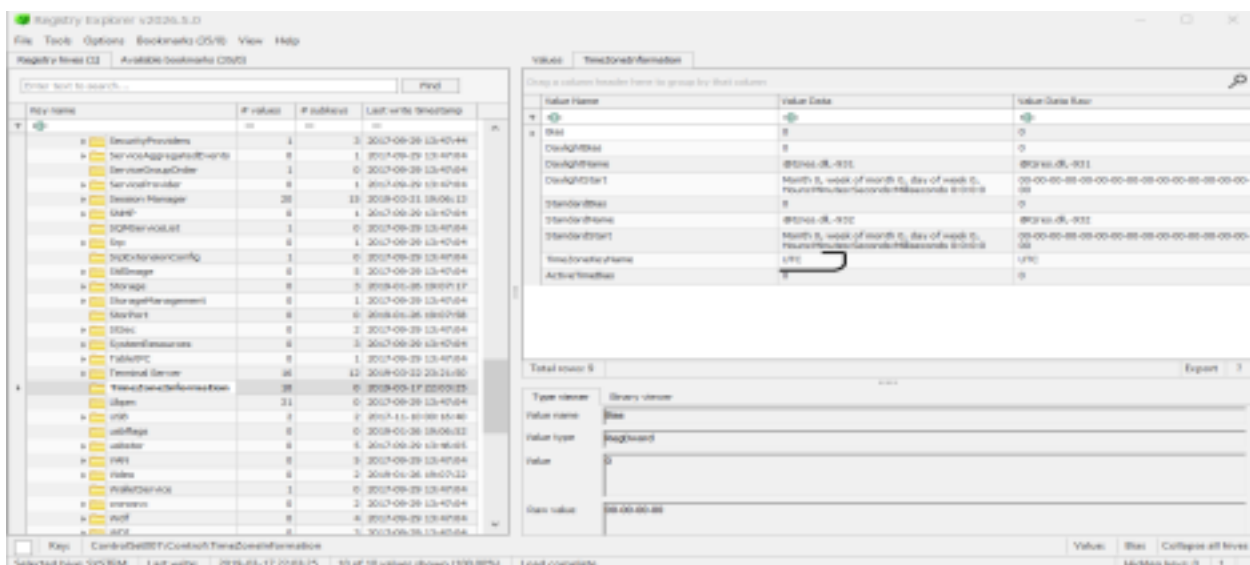
Method: The SYSTEM registry hive was loaded. Navigation path: ROOT > ControlSet001 > Control > ComputerName > ComputerName. The value ComputerName in the right panel showed TOTALLYNOTAHACK.



Q9 - Timezone (from SYSTEM hive)

Question: What is the machine's timezone? (Use the three-letter abbreviation) **Answer:** UTC

Method: Using the same SYSTEM hive, navigation path: ROOT > ControlSet001 > Control > TimeZoneInformation. The value TimeZoneKeyName in the right panel showed UTC.



Q14 - Password Last Changed (from SAM hive)

Question: When was the admin user password last changed?

Answer: 2019-03-22 23:22:01

Method: The SAM registry hive was loaded. Navigation path: ROOT > SAM > Domains > Account > Users > Names > Karen. Karen's account corresponded to the hex value 0x3E9 (1001 decimal), matching the folder 000003E9 under Users. Clicking on 000003E9 showed the Last write timestamp at the bottom of Registry Explorer as 2019-03-22 23:22:01, indicating when the password was last changed.

4.3 Findings from DB Browser for SQLite

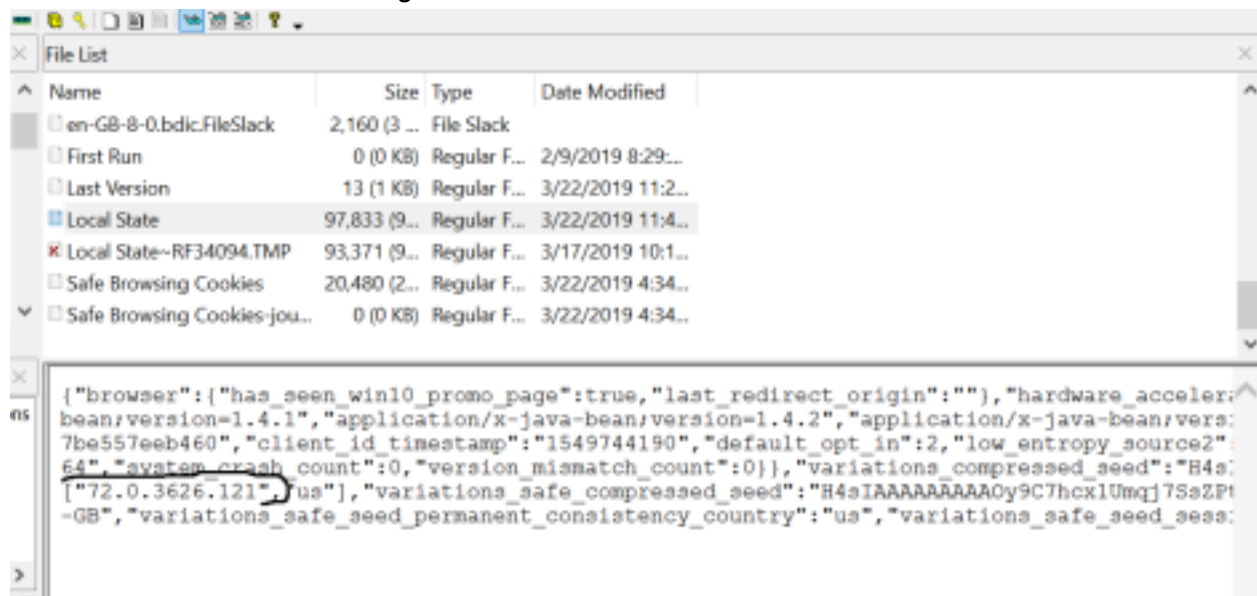
The Chrome History file was exported from Partition 2 > Users > Karen > AppData > Local > Google > Chrome > User Data > Default and opened in DB Browser for SQLite. SQL queries were run on the urls table to search for specific entries.

Q15 - Chrome Version

Question: What version of Chrome is installed on the machine?

Answer: 72.0.3626.121

Method: The Local State file was located at Partition 2 > Users > Karen > AppData > Local > Google > Chrome > User Data. Clicking on the file in FTK Imager revealed its contents in the preview panel. The text ["72.0.3626.121","us"] was visible, which is the Chrome version string stored in the Local State configuration file.



Q16 - Skype Download URL

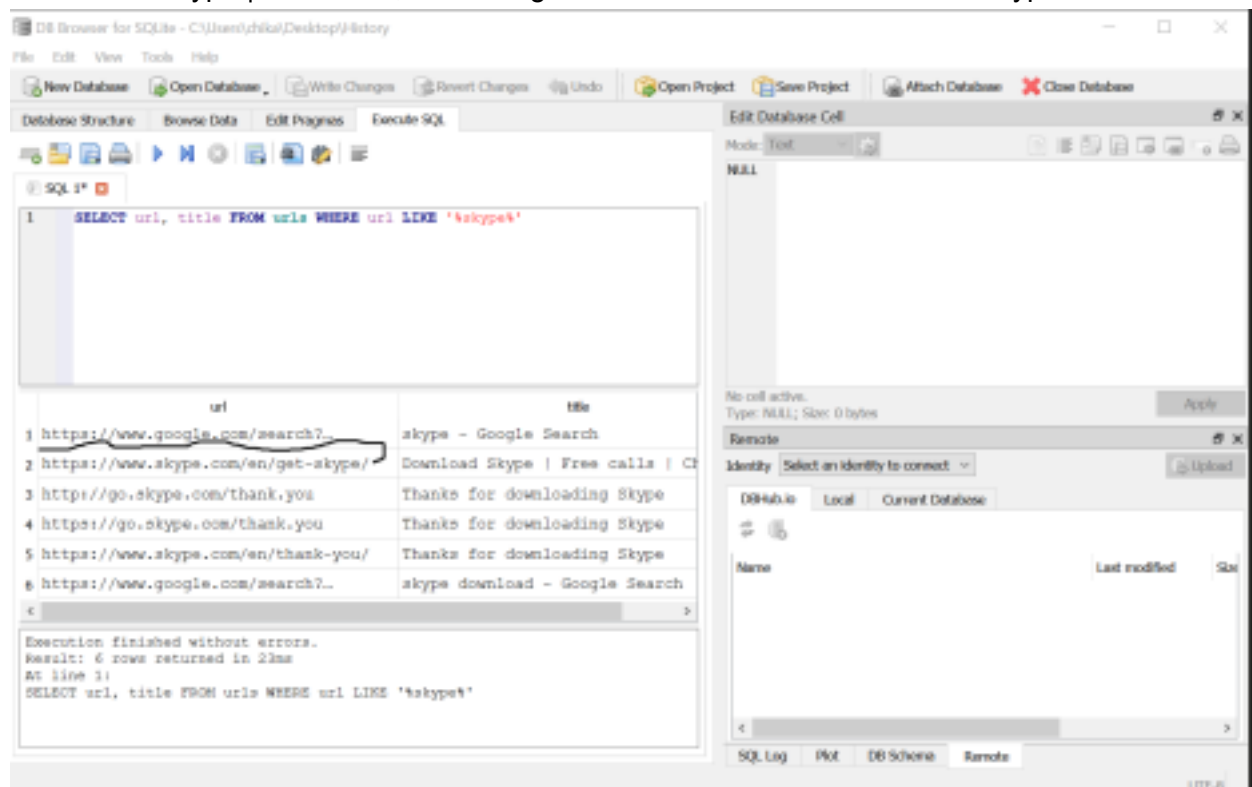
Question: What is the URL used to download Skype?

Answer: <https://www.skype.com/en/get-skype/>

Method: In DB Browser for SQLite, the Execute SQL tab was used. The following query was run:

```
SELECT url, title FROM urls WHERE url LIKE '%skype%'
```

Result: Row 2 of the results showed the URL <https://www.skype.com/en/get-skype/> with the title 'Download Skype | Free calls', confirming this is where Karen downloaded Skype.



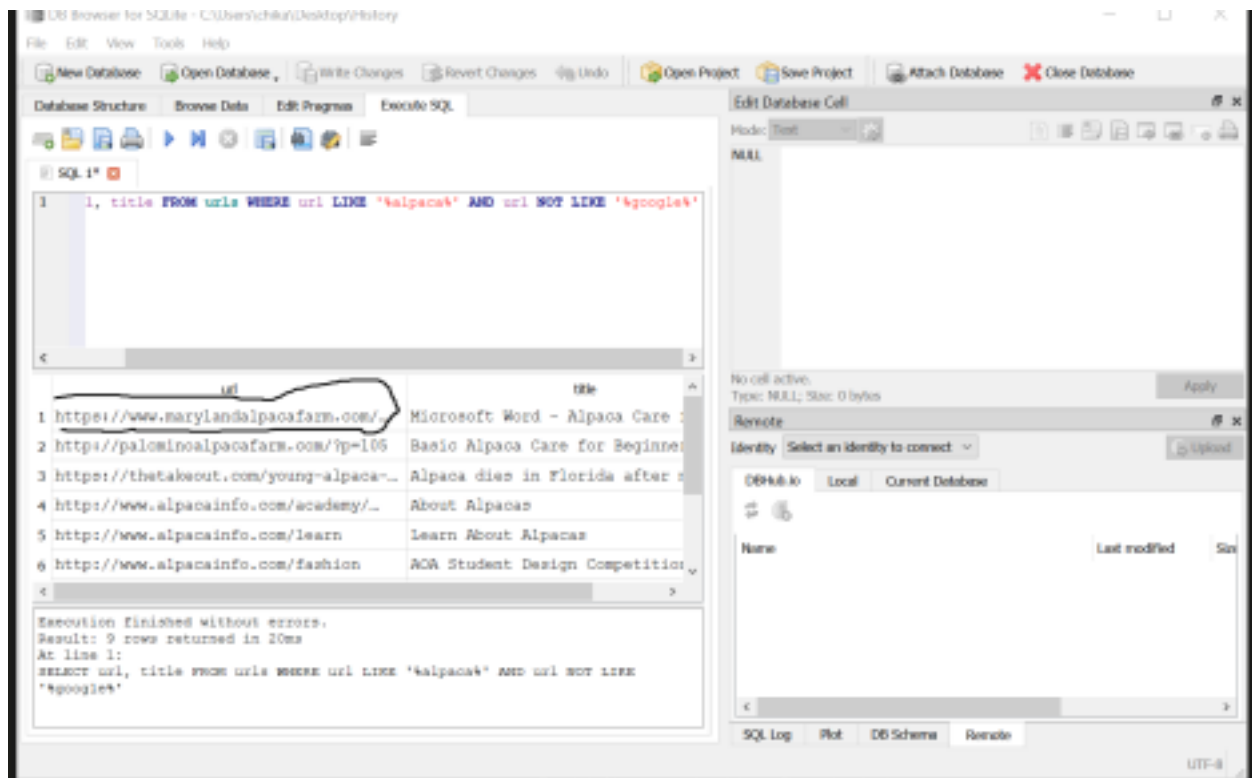
Q17 - Alpaca Care Website Domain

Question: What is the domain name of the website Karen browsed on Alpaca care that the file AlpacaCare.docx is based on?

Answer: www.marylandalpaca.com

Method: In DB Browser for SQLite, the Execute SQL tab was used. The following query was run to filter out Google search results and find the actual alpaca website Karen visited: `SELECT url, title FROM urls WHERE url LIKE '%alpaca%' AND url NOT LIKE '%google%'`

Result: Row 1 of the results showed the URL <https://www.marylandalpaca.com/> with the title 'Microsoft Word - Alpaca Care'. This confirms that Karen visited the Maryland Alpaca Farm website and used it as the basis for the AlpacaCare.docx document.



4.4 Findings from SysTools OST Viewer

The Outlook email database file klovespizza@outlook.com.ost was exported from Partition 2 > Users > Karen > AppData > Local > Microsoft > Outlook and opened in SysTools OST Viewer. The Inbox folder contained 19 emails which were examined.

Q5 - Zip Code

Question: What is the zip code of the administrator's post?

Answer: 19709

Method: An email in the inbox with subject 'Re: Job Needed, 19709' was found. The email body contained a Craigslist URL:

<https://vermont.craigslist.org/res/d/job-needed-19709/6815562978.html>. The number 19709 appears in both the subject line and the URL of Karen's own Craigslist post, identifying it as her zip code. US zip codes are 5-digit numbers, confirming 19709 is the zip code.

7066d7539fd30539e2e43b...	Job Offer "High paying	7066d7539fd30539e2e43b...	3/8/2019 3:16:10 AM	3/8/2019 3:16:28 AM	76
11fc12dac5d93bc396c4153...	Job Needed, 19709	11fc12dac5d93bc396c4153...	2/25/2019 9:29:12 PM	2/25/2019 9:29:21 PM	71
af46166350be3d8eb4536b...	Serious Employment Oppo...	af46166350be3d8eb4536b...	2/25/2019 2:18:19 PM	2/25/2019 2:18:33 PM	77

Normal Mail View | Attachments

Path : \\klovernsrta@outlook.com\et\PMRoot\Foot - Mailbox\PM SUBTREE\Inbox Date Time : 2/25/2019 9:29:12 PM

From : 11fc12dac5d93bc396c41532f3be1ab7@reply.craigslist.org

To : 11fc12dac5d93bc396c41532f3be1ab7@res.craigslist.org

Cc :

Bcc :

Subject : Job Needed, 19709

Attachment(s) :

Hello are there, how are you doing today? do you still need a job?

<https://vermont.craigslist.org/res/d/job-needed-19709/6815562978.html>

Original craigslist post:

<https://vermont.craigslist.org/res/6815562978.html>

About craigslist mail:

<https://craigslist.org/about/help/email-relay>

Q6 - TAAUSAI Contact Initials

Question: What are the initials of the person who contacted the admin user from TAAUSAI? **Answer:** M.S.

Method: The first email received from TAAUSAI (taausai@gmail.com) with the subject "Job Offer High paying" was opened. The email was signed by Micheal Scotch, who identified himself as working with The Alpaca Association of USA International (TAAUSAI). His initials are M.S.

Inbox						
From	Subject	To	Sent	Received	Size(KB)	
ms-amb@chromex.com	Onlineize your files with D...	klovernsrta@outlook.com	3/16/2019 6:44:55 PM	3/16/2019 6:44:57 PM	73	

Normal Mail View | Attachments

Path : \\klovernsrta@outlook.com\et\PMRoot\Foot - Mailbox\PM SUBTREE\Inbox Date Time : 3/8/2019 3:16:10 AM

From : 7066d7539fd30539e2e43ba9d21606@reply.craigslist.org

To : 7066d7539fd30539e2e43ba9d21606@res.craigslist.org

Cc :

Bcc :

Subject : Job Offer "High paying

Attachment(s) :

Hello Ms. Karen,

My name is Micheal Scotch and I work with The Alpaca Association of USA International (TAAUSAI). We here at the TAAUSAI are passionate about fighting for Alpaca animal rights in the United States of America and across the world, but we need your help!

We came across your Craigslist entry [here](#) and wanted to know if you'd be interested in a high paying technical job involving the use of computers. Don't worry about your skill level, we'll supply you with what you need.

Let us know what you think.

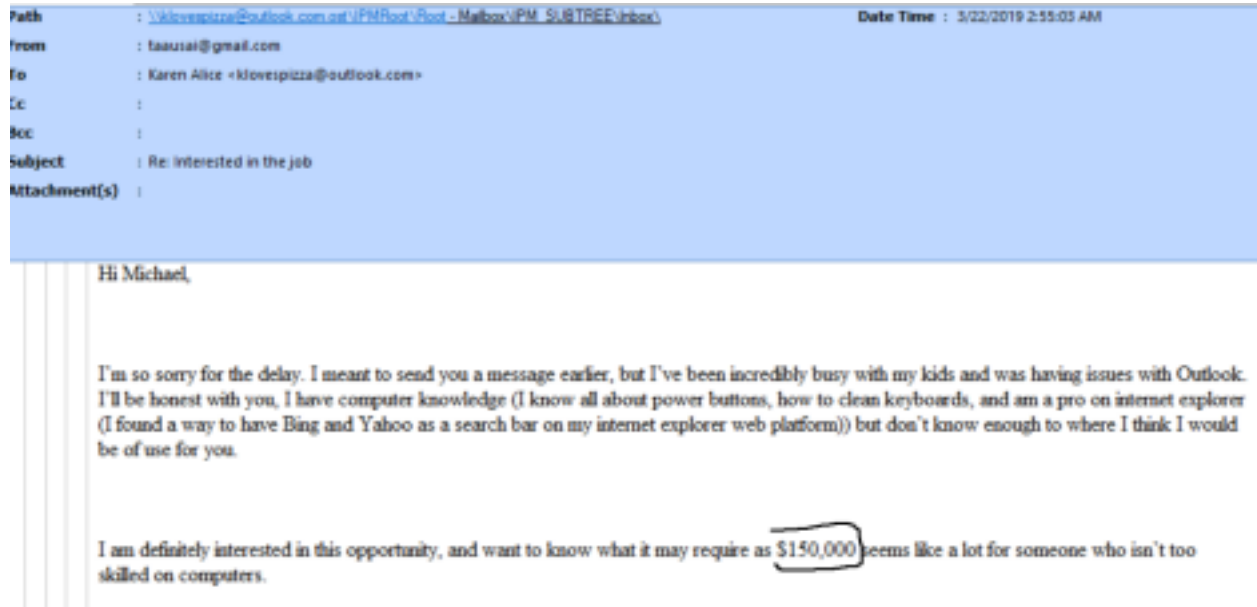
Best,
- Micheal Scotch

Q7 - Upfront Payment

Question: How much money was TAAUSAI willing to pay upfront?

Answer: \$150,000

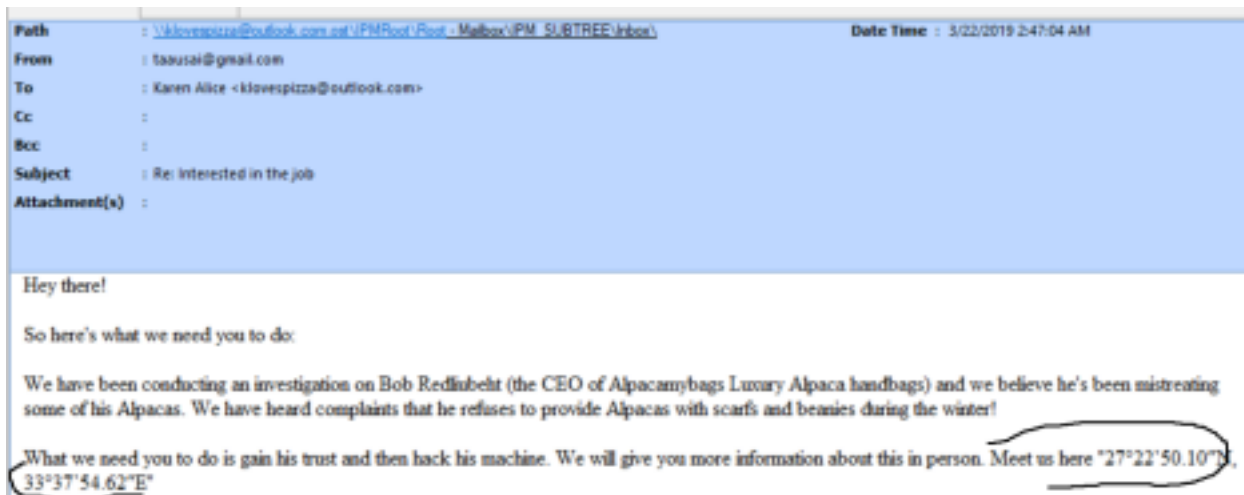
Method: In the email thread 'Re: Interested in the job' from taausai@gmail.com dated 3/17/2019 7:18:54 AM, Karen's reply stated: 'I am definitely interested in this opportunity, and want to know what it may require as \$150,000 seems like a lot for someone who isn't too skilled on computers.' This reveals that TAAUSAI offered \$150,000 as the upfront payment.



Q8 - Meeting Country

Question: What country is the admin user meeting the hacker group in? **Answer: Egypt**

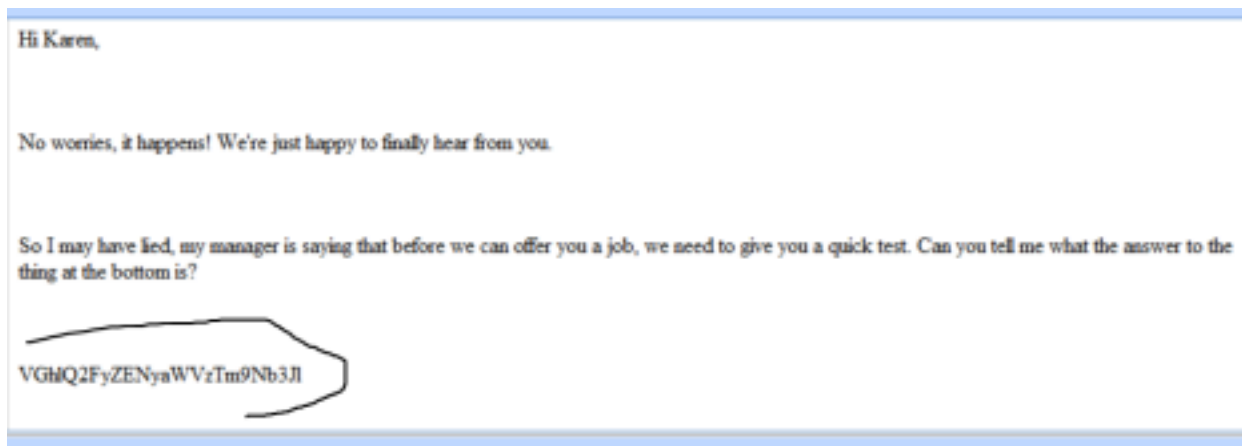
Method: In the email from taausai@gmail.com dated 3/22/2019 2:47:04 AM, Micheal Scotch wrote: 'What we need you to do is gain his trust and then hack his machine. We will give you more information about this in person. Meet us here 27 degrees 22 50.10 N, 33 degrees 37 54.62 E'. These GPS coordinates (27.38N, 33.63E) point to a location in Egypt, specifically near the Red Sea coast. This was verified using a coordinate mapping tool.



Q12 - Manager's Question Answer

Question: What is the answer to the question the company's manager asked Karen? **Answer:** TheCardCriesNoMore

Method: In the email from taausai@gmail.com dated 3/17/2019 7:44:12 AM, Micheal Scotch presented a test string: VGhIQ2FyZENyaWVzTm9Nb3Jl. This is a Base64 encoded string. When decoded, it translates to TheCardCriesNoMore. Karen confirmed the answer in her reply email where she typed: 'The answer is TheCardCriesNoMore'.



Path : \\klovespizza@outlook.com\root\IPMR\root\root - Mailbox\IPM - SUBTREE\inbox\ Date Time : 3/17/2019 7:44:12 AM
From : taausai@gmail.com
To : Karen Alice <klovespizza@outlook.com>
Cc :
Bcc :
Subject : Re: Interested in the job
Attachment(s) :

On Sun, Mar 17, 2019 at 2:34 AM Karen Alice <klovespizza@outlook.com> wrote:

Hi Michael,

The answer is TheCardCriesNoMore

-Karen

Q13 - Job Position Offered

Question: What is the job position offered to Karen? (3 words, 2 spaces in between) **Answer: Entry Level Cyber Security Analyst**

Method: In the email from taausai@gmail.com dated 3/22/2019 2:55:03 AM, Micheal Scotch wrote: 'The job position we think you will be an awesome fit for is an entry level cyber security analysts.' This identifies the job position offered as Entry Level Cyber Security Analyst.



4.5 File Metadata Finding from FTK Imager

Q10 - AlpacaCare.docx Last Accessed

Question: When was AlpacaCare.docx last accessed?

Answer: 3/17/2019 9:13:00 PM

Method: Windows automatically creates a shortcut (.LNK) file each time a document is opened. Navigation path: Partition 2 > Users > Karen > AppData > Roaming > Microsoft > Office > Recent. The file AlpacaCare.LNK was found here. Clicking on it and opening the Properties panel in FTK Imager revealed the MFT (Master File Table) timestamps. The Filename Date Accessed (MFT) showed 3/17/2019 9:13:00 PM, which is when AlpacaCare.docx was last accessed.



5. SUMMARY OF FINDINGS

The following table provides a complete summary of all 17 questions and their answers:

#	Question	Answer
Q1	What is the administrator's username?	Karen
Q2	What is the OS's build number?	16299
Q3	What is the hostname of the computer?	TOTALLYNOTAHACK
Q4	What messaging application was used?	Skype
Q5	What is the zip code of the administrator's post?	19709

Q6	What are the initials of the TAAUSAI contact?	M.S. (Micheal Scotch)
Q7	How much money was TAAUSAI willing to pay upfront?	\$150,000

Q8	What country is the admin meeting the hacker group in?	Egypt
Q9	What is the machine's timezone?	UTC
Q10	When was AlpacaCare.docx last accessed?	3/17/2019 9:13:00 PM
Q11	What letter is assigned to the second partition?	D
Q12	What is the answer to the manager's question?	TheCardCriesNoMore
Q13	What is the job position offered to Karen?	Entry Level Cyber Security Analyst
Q14	When was the admin user password last changed?	2019-03-22 23:22:01
Q15	What version of Chrome is installed?	72.0.3626.121
Q16	What is the URL used to download Skype?	https://www.skype.com/en/get-skype/
Q17	What is the domain of the Alpaca care website?	www.marylandalpacafarm.com

6. CONCLUSION

The forensic investigation of the Horcrux.ad1 image revealed that the administrator of the machine, Karen (email: klovespizza@outlook.com), was contacted by an organization calling itself The Alpaca Association of USA International (TAAUSAI) through a Craigslist job posting. The contact, Micheal Scotch, offered Karen a job as an Entry Level Cyber Security Analyst with a payment of \$150,000.

The investigation found evidence that Karen downloaded Skype (version 8.41.0.54) to communicate with the group. GPS coordinates provided in the emails point to a meeting location in Egypt. Karen was also asked to hack a target's machine as part of her job duties.

The machine ran Windows 10 Professional (Build 16299) with hostname TOTALLYNOTAHACK, set to UTC timezone. Karen's Chrome browser (version 72.0.3626.121) history showed she researched alpaca care on www.marylandalpaca.com, which formed the basis of the document AlpacaCare.docx, last accessed on 3/17/2019 at 9:13:00 PM.